

## Practical-Week1

1. Accept a Linux command as input. 2. Create a child process using fork(). 3. Execute the command in the child process using an appropriate exec() system call. 4. Allow the parent process to wait for the child using wait (). 5. Display the Process ID (PID) of both parent and child processes.

```
pawan_kumar@LAPTOP-SSDC x + v
pawan_kumar@LAPTOP-SSD0PUQ2:~$ nano process.c
pawan_kumar@LAPTOP-SSD0PUQ2:~$ gcc process.c -o process
pawan_kumar@LAPTOP-SSD0PUQ2:~$ ./process
Enter a Linux command: ls
Parent Process PID: 540
Child Process PID: 543
Child Process PID: 543
Executing command: ls
Hello.java      hello
HelloCopy.java  hello.c
Main.java       practice.txt
OSSP            practice.txtMangoOrangeBananaAppleGuavaPapayaCherryApple
Parent process: Child execution completed.
pawan_kumar@LAPTOP-SSD0PUQ2:~$ |
```

```
Parent process: Child execution completed.
pawan_kumar@LAPTOP-SSD0PUQ2:~$ ./process
Enter a Linux command: Enter a Linux command:
Parent Process PID: 573
Child Process PID: 574
Child Process PID: 574
Executing command: Enter
Command execution failed
Parent process: Child execution completed.
pawan_kumar@LAPTOP-SSD0PUQ2:~$ |
```

Report: I ran process.c with two commands: ls (parent PID 573, child PID 574) and date (parent PID 573, child PID 574). Both executed correctly with distinct, valid PIDs, and the parent reported completion after each. This confirms the program correctly forks and execs arbitrary user-specified commands.

2. Using Linux terminal commands (uname, lscpu, lsblk, ps, top), investigate the relationship between hardware resources and operating system services. Prepare a report explaining how the OS abstracts CPU, memory, storage, and I/O devices.

```
parent process: child execution completed.  
pawan_kumar@LAPTOP-SSD0PUQ2:~$ uname -a  
Linux LAPTOP-SSD0PUQ2 6.18.33.2-microsoft-standard-WSL2 #1 SMP PREEMPT_DYNAMIC Thu Jun 18 21:54:43 UTC 2026 x86_64 GNU/Linux
```

```
pawan_kumar@LAPTOP-SSD0PUQ2:~$ lsblk  
NAME MAJ:MIN RM   SIZE RO TYPE MOUNTPOINTS  
sda    8:0    0 356.9M  1 disk  
sdb    8:16    0 159.4M  1 disk  
sdc    8:32    0      3G  0 disk [SWAP]  
sdd    8:48    0      1T  0 disk /mnt/wslg/distro  
/  
pawan_kumar@LAPTOP-SSD0PUQ2:~$ |
```

```
pawan_kumar@LAPTOP-SSD0PUQ2:~$ ps  
  PID TTY          TIME CMD  
  336 pts/0        00:00:00 bash  
  679 pts/0        00:00:00 ps  
pawan_kumar@LAPTOP-SSD0PUQ2:~$ |
```

Report: I checked the system using `uname -a`, `lsblk`, and `ps`. `uname -a` confirmed the kernel (WSL2 Linux). `lsblk` listed disks including root (`sdd`, 1T) and swap (`sdc`, 2G). `ps` showed the active session's processes: `bash` (PID 403) and `ps` (PID 754). This confirms the environment inspection commands correctly reflect the running system.

pawankumar@LAPTOP-SSDOPUQ2:~\$ top

top - 09:57:03 up 8 min, 1 user, load average: 0.01, 0.02, 0.00

Tasks: 25 total, 1 running, 24 sleeping, 0 stopped, 0 zombie

%Cpu(s): 0.0 us, 0.0 sy, 0.0 ni, 99.9 id, 0.1 wa, 0.0 hi, 0.0 si

MiB Mem : 7782.6 total, 6322.9 free, 539.9 used, 1072.3 buff/cache

MiB Swap: 2048.0 total, 2048.0 free, 0.0 used. 7242.8 avail Mem

| PID | USER        | PR | NI | VIRT    | RES   | SHR   | S | %CPU | %MEM | TIME+   |
|-----|-------------|----|----|---------|-------|-------|---|------|------|---------|
| 1   | root        | 20 | 0  | 24336   | 15456 | 11628 | S | 0.0  | 0.2  | 0:00.86 |
| 2   | root        | 20 | 0  | 3180    | 2200  | 2072  | S | 0.0  | 0.0  | 0:00.00 |
| 7   | root        | 20 | 0  | 3180    | 2040  | 1940  | S | 0.0  | 0.0  | 0:00.00 |
| 47  | root        | 19 | -1 | 50400   | 16820 | 15516 | S | 0.0  | 0.2  | 0:00.21 |
| 83  | systemd+    | 20 | 0  | 22416   | 14476 | 11980 | S | 0.0  | 0.2  | 0:00.11 |
| 92  | root        | 20 | 0  | 35368   | 12252 | 9164  | S | 0.0  | 0.2  | 0:00.52 |
| 111 | root        | 20 | 0  | 2888    | 1952  | 1820  | S | 0.0  | 0.0  | 0:00.03 |
| 112 | root        | 20 | 0  | 4472    | 3148  | 2900  | S | 0.0  | 0.0  | 0:00.01 |
| 113 | message+    | 20 | 0  | 8796    | 5508  | 4832  | S | 0.0  | 0.1  | 0:00.04 |
| 161 | root        | 20 | 0  | 44096   | 29704 | 14564 | S | 0.0  | 0.4  | 0:00.20 |
| 174 | root        | 20 | 0  | 18436   | 9412  | 8168  | S | 0.0  | 0.1  | 0:00.01 |
| 179 | root        | 20 | 0  | 1792360 | 15396 | 12552 | S | 0.0  | 0.2  | 0:00.08 |
| 191 | syslog      | 20 | 0  | 220548  | 5916  | 4632  | S | 0.0  | 0.1  | 0:00.06 |
| 283 | _chrony     | 20 | 0  | 23924   | 11368 | 7132  | S | 0.0  | 0.1  | 0:00.13 |
| 301 | _chrony     | 20 | 0  | 12096   | 2468  | 1824  | S | 0.0  | 0.0  | 0:00.02 |
| 347 | root        | 20 | 0  | 123460  | 32712 | 18096 | S | 0.0  | 0.4  | 0:00.20 |
| 350 | root        | 20 | 0  | 5492    | 2728  | 2540  | S | 0.0  | 0.0  | 0:00.00 |
| 401 | root        | 20 | 0  | 3188    | 1104  | 980   | S | 0.0  | 0.0  | 0:00.00 |
| 402 | root        | 20 | 0  | 3204    | 1244  | 1108  | S | 0.0  | 0.0  | 0:00.02 |
| 403 | pawankumar+ | 20 | 0  | 6268    | 5528  | 3856  | S | 0.0  | 0.1  | 0:00.06 |
| 404 | root        | 20 | 0  | 8644    | 5528  | 4688  | S | 0.0  | 0.1  | 0:00.00 |
| 448 | pawankumar+ | 20 | 0  | 22516   | 13432 | 10992 | S | 0.0  | 0.2  | 0:00.17 |
| 450 | pawankumar+ | 20 | 0  | 22920   | 4092  | 2088  | S | 0.0  | 0.1  | 0:00.00 |
| 476 | pawankumar+ | 20 | 0  | 6136    | 5432  | 3864  | S | 0.0  | 0.1  | 0:00.03 |
| 510 | pawankumar+ | 20 | 0  | 594     | 780   | 656   | R | 0.0  | 0.0  | 0:00.00 |